SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 4

Managerial Accounting & Costing Systems (Job-Order & Process Costing)

From accounting for outsiders to accounting for managers: classifying costs, cost behavior, and two product-costing systems — job-order (unique orders) and process costing (continuous mass production). Includes a full worked solution to the Nusantara CraftWorks case.

Material

1 Cost classification depends on purpose

Classifications serve different purposes: direct vs indirect (tracing), DM/DL/MOH (manufacturing), product vs period (statements), variable/fixed/mixed (behavior), differential/opportunity/sunk (decisions).

`Prime = DM + DL` `Conversion = DL + MOH` Selling and administrative costs are period costs; product costs wait in inventory until sale.

2 Cost behavior and decision costs

Type	Total	Per unit
Variable	Proportional	Constant
Fixed	Constant in relevant range	Falls as activity rises
Mixed	$Y = a + bX$	Depends

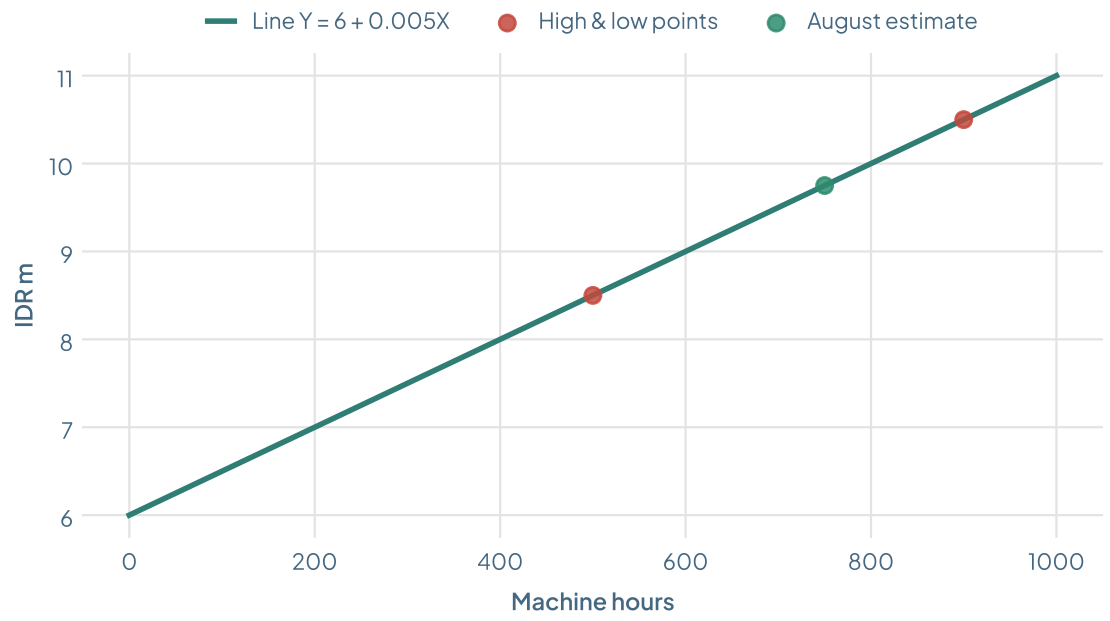
Cost drivers cause variable costs. Separate mixed costs by scattergraph, high-low ($b = \Delta\text{cost} \div \Delta\text{activity}$) or regression. Decision costs: differential and opportunity costs are relevant; **sunk costs are ignored**.

3 High-low drawn: separating fixed and variable cost

Deep dive

High-low uses the highest and lowest activity points to split mixed costs.

Power cost vs machine hours (high-low)



Intercept = fixed cost; slope = variable cost per hour.

1. b = (10.5m – 8.5m) ÷ 400 = IDR 5,000/hour.
2. a = IDR 6,000,000.
3. August (750 h) = IDR 9,750,000.

! **Weakness**

Only two points; regression uses all.

4 Job-order costing: for unique orders

Used for unique orders (aircraft, construction, consulting). The **job** is the cost object; the **job cost sheet** collects DM (requisitions), DL (time tickets) and applied overhead.

```
POHR = Estimated overhead ÷ Estimated allocation base; Applied = POHR × actual base
```

Example: $640k ÷ 160k DLH = $4/DLH; 8 DLH → $32. Average unit cost ($118) isn't incremental cost because it contains fixed overhead.

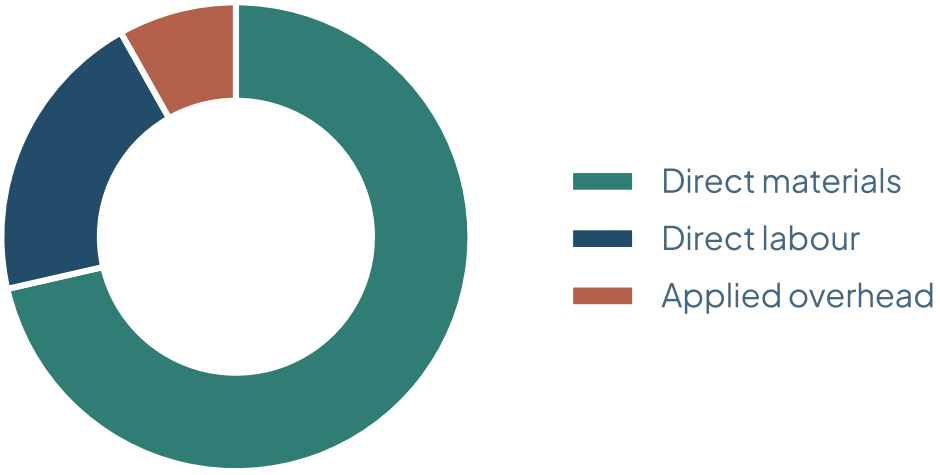
Flow: RM → WIP → FG (COGM) → COGS. COGM = total manufacturing costs + BWIP − EWIP; COGS = BFG + COGM − EFG ± under/overapplied overhead.

5 Job KR-118 cost and the job-order cost flow

Deep dive

Raw materials → WIP → Finished goods → COGS

Job KR-118 cost mix (IDR 58.8m)



IDR 58.8m	IDR 2.94m	~IDR 2.82m
Job total (20 chairs)	Average per chair	Incremental chair

6 Process costing: for uniform mass production

For homogeneous continuous production (beverages, cement, food). Costs accumulate by **department**; the production report is the key document.

Equivalent units: two half-done units = one complete unit. Weighted-average: EU = units transferred + EU in ending WIP; cost per EU = (BWIP cost + added cost) ÷ EU — separately for materials and conversion. Then cost ending WIP and transfers, and reconcile.

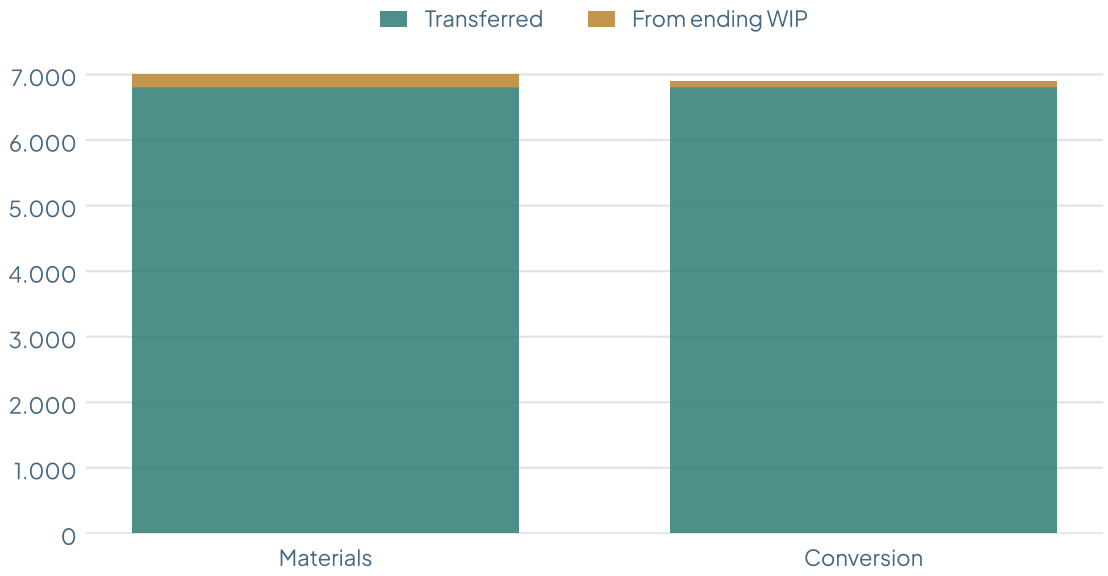
Operation costing is the hybrid: common processing plus customized finishing.

7 Equivalent units: why half-finished goods count partly

Deep dive

WIP counts partly: 500 units at 40% materials and 20% conversion = 200 and 100 equivalent units.

Equivalent units: materials vs conversion




```
Transferred 142.8m + ending WIP 3.3m = 146.1m ✔
```

8 Product vs period costs, and the cost of goods manufactured (COGM) schedule

Besides direct materials (DM), direct labour (DL) and manufacturing overhead (MOH), there are **non-manufacturing** costs: selling costs (advertising, commissions, shipping) and administrative costs (head-office salaries, accounting, legal). This split decides *when* a cost becomes an expense.

Period costs Selling and administrative costs. Expensed on the income statement in the period they occur, whether or not goods are sold.

Also remember two combinations from the slides: **prime cost = DM + DL** and **conversion cost = DL + MOH** (the cost of "converting" materials into product).

Example COGM and COGS schedule (IDR million):

Step	Calculation	Result
Raw materials used	Beginning 60 + Purchases 380 – Ending 50	390
+ Direct labour		250
+ Overhead applied		300
= Total manufacturing cost	390 + 250 + 300	940
+ Beginning WIP – Ending WIP	+80 – 70	
= Cost of goods manufactured (COGM)		**950**
+ Beginning finished goods – Ending finished goods	+120 – 100	
= Unadjusted cost of goods sold		970
+ Underapplied overhead	(see next section)	18
= Adjusted cost of goods sold (COGS)		**988**

```
COGM = DM used + DL + MOH applied + Beginning WIP - Ending WIP
COGS = Beginning finished goods + COGM - Ending finished goods
```

9 Job-order journal flow and under/overapplied overhead

In job-order costing, costs "flow" through the inventory accounts following the physical goods:

Raw Materials → Work in Process (WIP) → Finished Goods → Cost of Goods Sold

Event	Debit	Credit	Source document
Buy materials	Raw Materials	Accounts Payable	Purchase invoice
Use direct / indirect materials	WIP / Manufacturing Overhead	Raw Materials	Materials requisition form
Direct / indirect labour	WIP / Manufacturing Overhead	Salaries & Wages Payable	Employee time ticket
Actual overhead (utilities, rent, factory depreciation)	Manufacturing Overhead	Cash / Accumulated Depreciation	Bills, depreciation schedule
Apply overhead to jobs	WIP	Manufacturing Overhead	POHR × actual hours
Job completed	Finished Goods	WIP	Job cost sheet
Job sold	Cost of Goods Sold	Finished Goods	Sales invoice

Why is there an overhead difference? Overhead is applied to jobs using an *estimated* rate (POHR) set at the start of the year. At year-end, *actual* overhead is almost certainly different.

≡ **Example using the slide numbers**
POHR = ($200,000 + $2.75 × 160,000 DLH) ÷ 160,000 DLH = $640,000 ÷ 160,000 = **$4 per DLH**. Suppose actual hours are 158,000 DLH and actual overhead is $650,000.
Overhead applied = 158,000 × $4 = $632,000.
Difference = $650,000 – $632,000 = **$18,000 underapplied** → product costs were too low, so cost of goods sold is *increased* by $18,000. In the opposite case (overapplied), cost of goods sold is *reduced*.

A small difference is usually closed straight to cost of goods sold. A large one is allocated proportionally to WIP, Finished Goods and COGS so ending inventory is not misstated.

Operation costing (hybrid). Used when products share the same processes but use different materials — for example shoes made with different leathers, or clothing with different fabrics. Materials are costed per order/batch (as in job-order), while conversion costs are costed per operation/department (as in process costing).

Key takeaways

1. Cost classification depends on purpose.
2. Product costs → inventory → COGS; period costs expensed immediately.
3. Mixed cost Y = a + bX; separate via high-low or regression.
4. Ignore sunk costs; opportunity costs are relevant.
5. Job-order: cost per job; overhead applied via POHR.
6. Process: cost per department; equivalent units; reconcile.
7. Average unit cost ≠ incremental cost.

Case studies

Case & worked answers

Slide exercises — answers

MC1: wood direct/product; supervisor manufacturing overhead; commission period; head-office accounting administrative; factory depreciation indirect. MC2: A job-order, B process, C operation costing. MC3: POHR $4/MH; Job X $140; actual overhead isn't known in time. MC4: materials EU 480; conversion EU 200.

Case & worked answers

Pre-assignment: Nusantara CraftWorks — full solution

Part 1: items 1, 2, 4 direct manufacturing product costs; 3, 5, 6, 7 indirect manufacturing overhead; 8, 9 traceable to the order but selling (period) costs; 10 indirect administrative period cost. Prime cost IDR 276m; conversion IDR 240m. High-low: b = IDR 5,000/MH, a = IDR 6m → Y = 6,000,000 + 5,000X; August 750 MH → IDR 9.75m. Headboard: sunk cost IDR 60m; opportunity cost IDR 3m rent; relevant cost A = 18+9+2+3 = 32m, B = 32m — a tie, so qualitative factors decide; the machine's price is irrelevant.

Part 2: POHR = 720m ÷ 24,000 = IDR 30,000/DLH. KR-118: DM 42m + DL 12m + MOH 4.8m = 58.8m; IDR 2.94m per chair. Incremental cost ≈ 2.82m (excludes fixed overhead). COGM = 1,270m; unadjusted COGS 1,210m; underapplied overhead 10m → adjusted COGS 1,220m.

Part 3: EU materials 7,000, conversion 6,900; cost/EU 12,000 + 9,000 = 21,000; ending WIP 3.3m; transferred 142.8m; reconciles to 146.1m.

Closing: unique vs homogeneous output dictates the system; swapping would cause admin overload or mispricing. Semi-custom shelving → operation costing. High-low can split overhead into fixed and variable to set next year's POHR accurately.

Worked examples

Problem 1 **Easy** **Metode high-low**

Biaya pemeliharaan: bulan tersibuk 9.000 jam mesin = Rp52.000; bulan tersepi 4.000 jam = Rp32.000. Pisahkan biaya tetap dan variabel, lalu perkirakan biaya pada 7.000 jam.

1. `b = (Biaya tertinggi - Biaya terendah) ÷ (Aktivitas tertinggi - terendah) = 20.000 ÷ 5.000 = 4 per jam`
2. a = Biaya total – b × aktivitas = 52.000 – 4 × 9.000 = 16.000 (biaya tetap).
3. Y = 16.000 + 4 × 7.000 = 44.000.

Answer: Y = 16.000 + 4X; pada 7.000 jam biayanya Rp44.000.

Problem 2 Medium **Job-order costing**

Estimasi overhead setahun Rp600.000 dengan dasar 40.000 jam tenaga kerja langsung. Job #7: bahan langsung 8.000, tenaga kerja langsung 6.000 (400 jam), 500 unit. Harga jual = biaya + markup 50%. Berapa harga per unit?

1. `POHR = 600.000 ÷ 40.000 = 15 per jam`
2. Overhead dibebankan = 400 × 15 = 6.000.
3. Total biaya job = 8.000 + 6.000 + 6.000 = 20.000 → per unit = 20.000 ÷ 500 = 40.
4. Harga = 40 × 1,5 = **60**.

Answer: Biaya per unit 40; harga jual 60.

Problem 3 Challenging **Unit ekuivalen (rata-rata tertimbang)**

Departemen pencampuran: unit selesai 16.000; WIP akhir 4.000 unit (bahan 100%, konversi 25%). Total biaya konversi Rp340.000. Hitung unit ekuivalen bahan dan konversi, biaya konversi per unit ekuivalen, dan biaya konversi WIP akhir.

1. Bahan: 16.000 + 4.000 × 100% = **20.000** unit ekuivalen.
2. Konversi: 16.000 + 4.000 × 25% = **17.000** unit ekuivalen.
3. Biaya konversi per UE = 340.000 ÷ 17.000 = **20**.
4. Biaya konversi WIP akhir = 1.000 × 20 = 20.000; yang selesai = 16.000 × 20 = 320.000. Cek: 320.000 + 20.000 = 340.000 ✓

Answer: UE bahan 20.000, UE konversi 17.000; Rp20 per UE; WIP akhir (konversi) Rp20.000.

Problem 4 Easy **Prime cost and conversion cost**

This month's production costs (IDR million): direct materials 120, direct labour 80, manufacturing overhead 100. Compute prime cost and conversion cost.

1. `Prime cost = DM + DL = 120 + 80 = 200`

2. `Conversion cost = DL + MOH = 80 + 100 = 180`

3. Direct labour sits in both, so prime + conversion (380) ≠ total manufacturing cost (300). Do not add them together.

Answer: Prime cost IDR 200 million; conversion cost IDR 180 million.

Problem 5 Medium **COGM and COGS schedule**

(IDR million) Opening raw materials 30, purchases 200, closing raw materials 40. Direct labour 150. Overhead applied 180. Opening WIP 50, closing WIP 60. Opening finished goods 70, closing finished goods 90. Compute COGM and COGS.

1. Raw materials used = 30 + 200 – 40 = 190.
2. Total manufacturing cost = 190 + 150 + 180 = 520.
3.
```
COGM = 520 + opening WIP 50 - closing WIP 60 = 510
```
4.
```
COGS = Opening FG 70 + COGM 510 - Closing FG 90 = 490
```

Answer: COGM IDR 510 million; COGS IDR 490 million (before the overhead adjustment).

Problem 6 Medium **Under/overapplied overhead**

POHR = IDR 600,000 ÷ 40,000 hours = IDR 15 per direct labour hour. At year-end, actual hours are 42,000 and actual overhead is IDR 610,000. Compute overhead applied, the difference, and its effect on COGS.

1. `Overhead applied = POHR × Actual hours = 15 × 42,000 = IDR 630,000`
2. Difference = Actual – Applied = 610,000 – 630,000 = –20,000 → overhead is **overapplied** by IDR 20,000.
3. Product costs were recorded too high, so COGS is *reduced* by IDR 20,000 (if a small difference is closed to COGS).

Answer: Overapplied by IDR 20,000 → COGS falls by IDR 20,000.

Practice questions

Try answering first. The answer key is below each question.

1. High-low: 900 h = 10.5m, 500 h = 8.5m. Variable cost per hour…

A. 2,000
B. 5,000
C. 11,667
D. 17,000

Answer: B. 2m ÷ 400 = 5,000.

2. POHR 30,000/DLH, job uses 160 DLH. Applied overhead…

A. 480k
B. 4.8m
C. 12m
D. 48m

Answer: B. 4.8m.

3. A continuous bottled-tea producer should use…

A. Job-order
B. Process costing
C. None
D. Operation only

Answer: B. Uniform output → process costing.

---

4. A machine bought 3 years ago for 60m is, for today's decision…

A. Relevant
B. Opportunity cost
C. Sunk cost — irrelevant
D. Variable

Answer: C. Already spent.

---

5. 500 WIP units 20% complete for conversion equal…

A. 20
B. 100
C. 400
D. 500

Answer: B. 100.

Sources

- -P3MD- Week 4 MK02.pdf
- Pre-Assignment MK02 Week 4 Nusantara CraftWorks Case Study.pdf